Build and Configure a Small Office Network

Task 1 — Assign Static IP Addresses to Five Devices

IP Addressing Plan

Device	IP Address	Subnet Mask	Default Gateway
Laptop 1	192.168.10.10	255.255.255.0	192.168.10.1
Laptop 2	192.168.10.11	255.255.255.0	192.168.10.1
Desktop 1	192.168.10.12	255.255.255.0	192.168.10.1
Desktop 2	192.168.10.13	255.255.255.0	192.168.10.1
Desktop 3	192.168.10.14	255.255.255.0	192.168.10.1

Configuration

- Assign the five static IPv4 addresses shown above to the respective devices.
- Use subnet mask 255.255.255.0 on all five devices so they belong to the same /24 subnet.
- Configure 192.168.10.1 as the default gateway.

Connectivity Verification

```
ipconfig
ping 192.168.10.11
ping 192.168.10.12
ping 192.168.10.13
ping 192.168.10.14
```

Completed Result

- All five devices are in the same subnet and have unique IP addresses.
- Successful ping replies between the devices confirm local LAN connectivity.

Task 2 — Configure a DHCP Server

DHCP Server Configuration

Setting	Value
DHCP Server IP	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	192.168.10.2
Start IP Address	192.168.10.100
Maximum Users	51

Procedure in Cisco Packet Tracer

- Place a Server, Switch and at least three client PCs in the topology.
- Connect the server and clients to the switch using copper straight-through cables.
- Assign the server the static address 192.168.10.2/24.
- Open Server → Services → DHCP and turn the DHCP service ON.
- Enter the pool settings shown above and save the configuration.
- On each client, open Desktop → IP Configuration and select DHCP.

Completed Client Results

Device	DHCP Address	Mask	Gateway
DHCP Client 1	192.168.10.100	255.255.255.0	192.168.10.1
DHCP Client 2	192.168.10.101	255.255.255.0	192.168.10.1
DHCP Client 3	192.168.10.102	255.255.255.0	192.168.10.1

Verification

- Each client receives a unique address from the configured DHCP range.
- The addresses are in the correct 192.168.10.0/24 network.
- The clients can use 192.168.10.1 as their default gateway and 192.168.10.2 as DNS.

Task 3 — Configure a Local DNS Service

DNS Configuration

- DNS server: 192.168.10.2
- Custom hostname: musicplayer.local
- Local web/device interface: 192.168.10.50

DNS Record

```
musicplayer.local    A    192.168.10.50
```

Packet Tracer Procedure

- Use the DNS service on the server at 192.168.10.2.
- Turn DNS service ON and create an A record for musicplayer.local.
- Set the record address to 192.168.10.50.
- Ensure the client devices use 192.168.10.2 as their DNS server, either manually or through DHCP.

Verification

```
nslookup musicplayer.local 192.168.10.2
ping musicplayer.local
```

- Expected DNS result: musicplayer.local resolves to 192.168.10.50.
- A successful ping by hostname confirms that the client can resolve the custom name and reach the configured local address.

Completed Result

- The custom hostname musicplayer.local successfully maps to 192.168.10.50 through the local DNS server.

Task 4 — Create and Test Two VLANs

VLAN Plan

VLAN	Name	Purpose	Example Devices
10	Workstations	Office laptops/desktops	PC1, PC2
20	Smart Devices	Smart/IoT devices	Smart TV, Smart Device

Switch Configuration

```
enable
configure terminal
vlan 10
name Workstations
vlan 20
name Smart_Devices
interface range fa0/1-2
switchport mode access
switchport access vlan 10
interface range fa0/3-4
switchport mode access
switchport access vlan 20
end
show vlan brief
```

Expected VLAN Membership

- FastEthernet0/1 and FastEthernet0/2 belong to VLAN 10 — Workstations.
- FastEthernet0/3 and FastEthernet0/4 belong to VLAN 20 — Smart Devices.

Connectivity Test

- Devices in VLAN 10 can communicate with other devices in VLAN 10 when their IP configuration is correct.
- Devices in VLAN 20 can communicate with other devices in VLAN 20 when their IP configuration is correct.
- With no inter-VLAN routing configured, a VLAN 10 device cannot communicate directly with a VLAN 20 device.
- Inter-VLAN communication would require a Layer 3 switch or router-on-a-stick configuration with appropriate gateway interfaces.

Conclusion

- VLANs provide logical network segmentation. Same-VLAN communication works, while communication between separate VLANs is blocked until routing is configured.

Task 5 — Simulate Internet Access with NAT

Network Design

- LAN interface of Router: 192.168.10.1/24
- WAN interface of Router: 203.0.113.2/30
- Simulated ISP/external server: 203.0.113.1
- Inside LAN example client: 192.168.10.100

Router NAT Configuration

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0
ip nat inside
no shutdown
exit
interface gigabitEthernet 0/1
ip address 203.0.113.2 255.255.255.252
ip nat outside
no shutdown
exit
access-list 1 permit 192.168.10.0 0.0.0.255
ip nat inside source list 1 interface gigabitEthernet 0/1 overload
ip route 0.0.0.0 0.0.0.0 203.0.113.1
end
```

Testing

```
ping 203.0.113.1
show ip nat translations
show ip nat statistics
```

Expected NAT Operation

- The LAN device sends traffic using its private address, for example 192.168.10.100.
- The router translates the source address to its WAN address 203.0.113.2 using PAT/NAT overload.
- The simulated external server receives the translated traffic and returns the response.
- The router translates the return traffic back to the internal client.

Completed Result

- LAN clients can reach the simulated external server through the router.
- NAT allows multiple private LAN devices to share the router's external address while accessing the simulated Internet.

Detailed Network Setup and Configuration Report

Overall Topology

- End Devices → Access Switch → Router → Simulated ISP/External Server.
- DHCP and DNS services are provided by the local server at 192.168.10.2.
- The router provides the LAN default gateway and performs NAT for simulated Internet access.
- VLAN 10 is used for workstations and VLAN 20 is used for smart devices.

Configuration Summary

Service/Feature	Configuration	Verification
Static IP	192.168.10.10–192.168.10.14	Ping between devices
DHCP	192.168.10.100–192.168.10.150	ipconfig / DHCP settings
DNS	musicplayer.local → 192.168.10.50	nslookup / ping hostname
VLAN 10	Workstations	show vlan brief
VLAN 20	Smart Devices	show vlan brief
NAT	192.168.10.0/24 → WAN interface	show ip nat translations

Troubleshooting Notes

- If a device cannot ping another device in the same VLAN, check the IP address, subnet mask, cable/link status, VLAN assignment and local firewall settings.
- If a client receives no DHCP address, check the DHCP service, pool range, server connectivity and switch port.
- If a hostname does not resolve, check the client's DNS server address and the DNS A record.
- If VLAN 10 cannot communicate with VLAN 20, verify whether inter-VLAN routing has been configured; without a Layer 3 gateway, this behavior is expected.
- If Internet simulation fails, check the router's WAN interface, default route, NAT inside/outside settings and access list.

Final Result

- The completed design provides static addressing, automatic DHCP addressing, local DNS resolution, VLAN-based segmentation, simulated Internet connectivity through NAT, and a documented troubleshooting procedure.